

Body of Knowledge

Machine Learning Expert Group

February 2026

Brief discussion

Machine learning (ML) is a branch of artificial intelligence that focuses on enabling systems to learn patterns directly from data using statistical and computational methods, rather than relying on explicitly programmed rules [2]. The outcome of this learning process is a model that encodes regularities in the data and can be used either to explain observed data patterns, such as human behaviour, or to generate predictions for new, unseen inputs. In practice, ML underpins a wide range of applications in socio-technical systems, including anomaly detection, recommender systems, trend detection, and forecasting.

Most ML workflows distinguish between two phases. During **training**, the model is exposed to data and iteratively updates its internal parameters in order to learn underlying patterns and improve its performance with respect to a specified objective. After training, the trained model is deployed in **inference** mode, where it applies the acquired knowledge to produce outputs for previously unseen inputs.

Machine learning includes many learning strategies and model families (see Figure 1). Below we provide a representative (although not exhaustive) overview.

Three learning strategies

The three learning strategies below are best distinguished by the **training signal** available during learning: labelled targets (supervised), structure in unlabelled data (unsupervised/self-supervised), or rewards from interaction over time (reinforcement learning). The same model family (for example, decision trees or neural networks) can typically be used within multiple learning strategies.

Supervised learning

In supervised learning, training data consist of input–output examples. The goal is to learn a mapping from inputs to targets (i.e. outputs) that generalises well to new, unseen cases. Two common supervised tasks are:

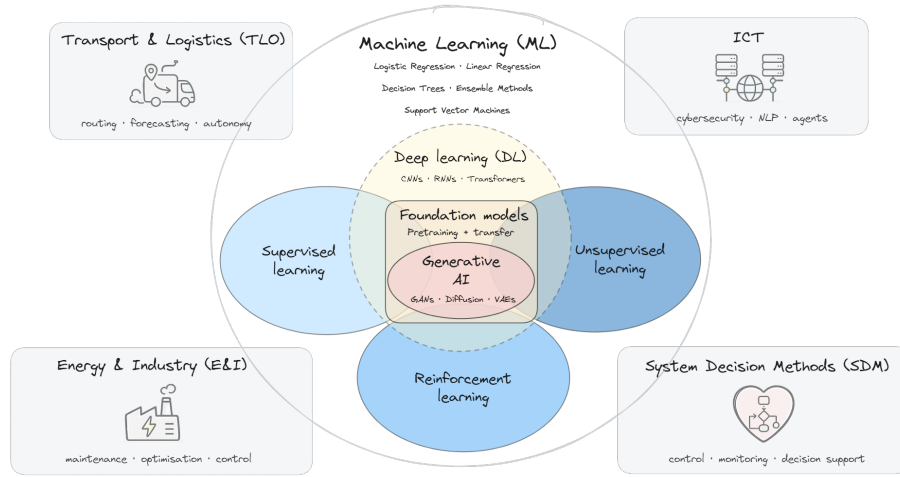


Figure 1: Venn diagram illustrating the relationship between machine learning strategies, model families, and emerging topics.

- **Regression:** predicting a continuous quantity. Training typically penalises the difference between predicted and measured values, e.g. using a mean squared error.
- **Classification:** predicting a discrete category. Training typically penalises incorrect class assignments, often using probability-based objectives, such as a log-likelihood.

Supervised learning is widely used when reliable labels are available, and performance can be validated against measured outcomes.

Unsupervised learning (including self-supervised learning)

In unsupervised learning, only inputs are observed, without explicit labels. The goal is to discover structure in the data, such as groups, low-dimensional representations, or regularities that can support downstream tasks. Common examples include:

- **Clustering:** grouping samples so that points within a group are more similar to each other than to points in other groups.
- **Representation learning:** learning compact features (embeddings) that capture important variation in the data and make downstream tasks easier.

A widely used modern variant is **self-supervised learning**, where the system creates its own learning targets from the input (for example predicting missing parts of an input or the next element in a sequence). This is often grouped under unsupervised learning in practice because it does not require manual labels.

Reinforcement learning (RL)

Reinforcement learning studies sequential decision-making. An *agent* repeatedly observes a situation (the *state*) and chooses an *action* in an *environment*. The environment responds by transitioning to a new situation and providing a numerical *reward* signal indicating how good the outcome was. The agent's goal is to learn a *policy* (a rule for choosing actions from states) that maximises long-term reward, not just immediate gains. RL is most appropriate when decisions affect future outcomes, feedback can be delayed, and the system must trade off *exploration* (trying actions to learn) and *exploitation* (using what currently seems best).

Model families

Model families describe **how** the mapping from inputs to outputs is algorithmically represented. They are not learning strategies. The same family can be trained using supervised, unsupervised/self-supervised, or reinforcement learning, depending on the training signal and objective.

Traditional machine learning

Traditional machine learning refers to widely used model classes that are typically trained on structured (tabular) data and may require explicit feature engineering. Representative examples include:

- **Linear regression and logistic regression:** simple and efficient baselines that can be strong when relationships are close to linear or when features are well designed.
- **Decision trees and ensemble methods:** Trees are easy to interpret at small depth; ensembles (random forests and gradient-boosted trees) often deliver strong performance on tabular data and handle mixed feature types well.
- **Random Forest:** An ensemble of decision trees trained on bootstrapped samples with random feature selection. By averaging across many trees, random forests reduce variance and typically achieve robust performance on tabular data with limited tuning.
- **Gradient Boosting:** An ensemble technique that builds trees sequentially, where each new tree focuses on correcting the errors of the previous ensemble. Gradient-boosted trees often achieve state-of-the-art performance on structured datasets and can model complex nonlinear relationships.

These methods are frequently preferred when datasets are moderate in size, interpretability is important, and the input representation is already meaningful.

Deep learning

Deep learning refers to using multi-layer neural networks to learn complex input–output mappings and internal representations. Rather than relying on hand-crafted features, deep networks can learn useful features directly from high-dimensional data such as images, text, audio, or wearable-sensor signals [1]. Common architecture families include:

- **Convolutional neural networks (CNNs):** well suited to data with local structure (for example, images and some sensor signals).
- **Recurrent neural networks (RNNs) and variants:** designed for sequential data such as time series and text.
- **Transformers:** attention-based models that scale well and are widely used in language and increasingly in vision, audio, and time series.

Two emerging topics

The topics below are not additional learning strategies. They describe emerging system-level patterns that build on the learning strategies and model families above.

Foundation models

Foundation models are large-capacity models trained on broad and diverse data to learn general-purpose representations. They are typically developed in two stages. First, the model is *pretrained* at scale, most often using self-supervised objectives that do not require manual labels (for example, predicting missing parts of an input or the next element in a sequence). Second, the pretrained model is *adapted* to a specific downstream task, for example, by fine-tuning on task-specific data, training lightweight adapters, or using the model as a feature extractor. The motivation is that large-scale pretraining yields reusable representations, reducing the amount of labelled data and task-specific engineering needed for new applications.

Generative AI

Generative AI refers to models that can produce new outputs such as text, images, audio, code, or time-series that resemble (or are consistent with) patterns seen during training. Many generative systems are built from foundation models, but generative capability is distinct from the foundation-model training recipe. In practice, generative systems are often guided by prompts and may be refined with task-specific data and preference-based feedback to improve usefulness and reduce harmful outputs. Common families of generative models include autoregressive models, variational autoencoders (VAEs), diffusion models, and generative adversarial networks (GANs).

Explainable AI (XAI)

Explainable AI (XAI) are methods that help users understand, validate, and appropriately use ML models, especially in safety-, cost-, or compliance-critical settings. XAI typically addresses questions such as: *Why did the model produce this output? Which inputs influenced the decision most? When is the model likely to be wrong?* Common approaches include:

- **Interpretable-by-design models:** choosing simpler model classes when they meet performance requirements (for example linear models or shallow decision trees).
- **Post-hoc explanations:** producing explanations for a trained model, such as feature attribution (global importance and per-prediction explanations) or example-based explanations that retrieve similar cases.
- **Counterfactual analysis:** describing what changes to the input would change the output, which is often useful for debugging and stakeholder communication.
- **Uncertainty and monitoring:** augmenting predictions with confidence estimates, detecting out-of-distribution inputs, and monitoring performance drift after deployment.

Applicability

ICT cluster applications

ICT applications include digital service operations, cyber/telemetry security, and governance in socio-technical contexts. To date, however, ML in this unit has been primarily used in healthcare-oriented ICT applications, particularly in nanomedicine, drug delivery, and interpretable cancer-treatment modelling.

Illustrative examples:

- **AI-assisted nanocarrier design and drug delivery:** Machine learning can support the design, synthesis, and optimization of nanoparticle formulations, for example by improving drug encapsulation, release kinetics, targeted delivery, and the ability to overcome physiological barriers [6].
- **Interpretable personalized oncology:** ML can also be combined with mechanistic modelling to support personalized cancer treatment. For example, by combining Kolmogorov-Arnold Networks with evolutionary game theory it is possible to obtain more interpretable and clinically usable models of cancer progression and treatment response [7].

Transport and Logistics cluster applications

In the Transport and Logistics cluster, ML is frequently used to model, predict, and explain behavioural and operational phenomena in complex mobility systems. Applications typically combine behavioural modelling traditions with modern data-driven techniques. **Illustrative examples:**

- **Inference of choice behaviour:** Machine learning and hybrid choice models can be used, e.g., to investigate (heterogeneous) preferences, such as the distribution of the Value-of-Travel-Time (VTT) [9] and decision rules [14], and explore nonlinearities in preferences [4]. These approaches increase functional flexibility while retaining behavioural interpretability.
- **Investigation of choice behaviour with visual stimuli:** The integration of computer vision into choice modelling enables the analysis of visual environmental preferences. Recent studies explain cycling route and residential location choices using street-level imagery [8, 13], linking visual features to behaviour.
- **Assistance with modelling workflows:** Specification of choice models is often a laborious and time-consuming task. By formulating this process as a sequential decision-making problem, reinforcement learning (RL) agents can support the exploration of candidate model structures [11]. In addition, large language models (LLMs) have been explored as tools for generating candidate specifications and structuring model comparisons [12].

Across these examples, ML is not only used for prediction, but also to enhance behavioural insight, and bridge data-rich environments (e.g. imagery, sensor streams) with established transport modelling frameworks.

Energy and Industry cluster applications

Energy and Industry (E&I) applications include forecasting, market behavior, and optimization under uncertainty, where ML can either (i) improve predictive inputs to optimization or (ii) improve decision making in complex systems.

Systems Decision Methods cluster applications

The System Decision Methods (SDM) unit focuses on developing and applying methods for socio-technical systems, and on connecting these methods to the department’s application domains as well as emerging areas such as healthcare and resilience. In this context, ML is currently used by the unit in computational biomechanics. **Illustrative examples:**

- **Foundation Models for Human Movement Analysis:** learning reusable representations from large, diverse wearable and movement datasets to improve generalization across patients, sites, and measurement setups [5].

- **Adaptive Control of Assistive Devices:** personalizing and adapting control strategies for assistive technologies (e.g., exoskeletons/prosthesis) using continuous sensor feedback and safety/comfort constraints [10].
- **Decision Support in Parkinson’s Disease:** integrating wearable/clinical data to support monitoring and assessment, and to inform clinically meaningful decisions (e.g., symptom detection, progression tracking, or intervention planning) [15].

Educational offer at TPM

- **TPM034A — Machine Learning for Socio-technical Systems.** Equips students in the socio-technical domain with a deep understanding of machine learning and prepares them for ML-related challenges in practice (requires prior linear algebra, statistics/probability, data structures, and basic Python).
- **SEN182A — Human-centered Care Interventions.** Focuses on health and care as a sensitive AI application domain: students analyze weaknesses of current standard-of-care and design/prototype a human-centered, value-driven, safe AI-enabled solution (e.g., decision support, recommender systems, information retrieval, ML, generative/explainable AI).

Pitfalls/debate

Machine learning, despite its successes, raises several methodological and societal concerns [3]. Models often rely on large datasets that may be biased which could lead to unfair or unreliable outcomes. Many emerging approaches (particularly in deep learning) lack transparency, making it difficult to interpret decisions or establish causal understanding. Performance can degrade under distribution shifts, where deployment conditions differ from training data. Moreover, there is ongoing debate about reproducibility, data governance, and environmental costs of large-scale training.

Key references

- [1] Ian Goodfellow. “Deep learning”. In: 800 (2016).
- [2] Christopher M. *Pattern Recognition and Machine Learning*. en. 1st ed. Information Science and Statistics. New York, NY: Springer, Aug. 2006.
- [3] Nestor Maslej et al. “Artificial Intelligence Index Report 2025”. In: *arXiv [cs.AI]* (Nov. 2025).

- [4] Nicolas Salvadé and Tim Hillel. “RUMBoost: Gradient boosted random utility models”. In: *Transportation Research Part C: Emerging Technologies* 170 (2025), p. 104897.

Key articles from TPM researchers

- [5] Vida Adeli et al. “Care-PD: A Multi-Site Anonymized Clinical Dataset for Parkinson’s Disease Gait Assessment”. In: *The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025)*. Oct. 2025. URL: <https://neurips2025.care-pd.ca/>.
- [6] Sepinoud Azimi. “Intelligent nanoparticle design: Unlocking the potential of AI for transformative drug delivery”. en. In: *Curr. Opin. Biomed. Eng.* 36.100625 (Dec. 2025), p. 100625.
- [7] Sepinoud Azimi, Louise Spekking, and Kateřina Staňková. “Kolmogorov-Arnold Networks and Evolutionary Game Theory for more personalized cancer treatment”. In: *arXiv [cs.LG]* (Jan. 2025).
- [8] Sander van Cranenburgh and Francisco Garrido-Valenzuela. “Computer vision-enriched discrete choice models, with an application to residential location choice”. In: *Transportation Research Part A: Policy and Practice* 192 (2025), p. 104300.
- [9] Sander van Cranenburgh and Marco Kouwenhoven. “An artificial neural network based method to uncover the value-of-travel-time distribution”. In: *Transportation* 48.5 (2021), pp. 2545–2583.
- [10] Diwas Lamsal et al. “*RevalExo: A Multimodal Dataset and Benchmark for Locomotion Mode Recognition Across Healthy and Clinical Populations*”. 2026. DOI: 10.48804/OWJ0ID. URL: <https://revalexo.github.io/>.
- [11] Gabriel Nova, Stephane Hess, and Sander van Cranenburgh. “Delphos: A reinforcement learning framework for assisting discrete choice model specification”. In: *arXiv preprint arXiv:2506.06410* (2025).
- [12] Georges Sfeir et al. “Can large language models assist choice modelling? Insights into prompting strategies and current models capabilities”. In: *arXiv preprint arXiv:2507.21790* (2025).
- [13] Roosmarijn Willemijn Terra et al. “Understanding cycling route choice behaviour through street-level images and discrete choice models”. In: *Available at SSRN 5473360* ().
- [14] Sander Van Cranenburgh and Ahmad Alwosheel. “An artificial neural network based approach to investigate travellers’ decision rules”. In: *Transportation Research Part C: Emerging Technologies* 98 (2019), pp. 152–166.
- [15] Po-Kai Yang et al. “Deep learning for Freezing of Gait assessment using inertial measurement units: A multicentre study”. In: *medRxiv* (June 2025). URL: <https://aidfog.be/>.